Model Tuning Report - Week 1, Day 4

UCI Adult (Census Income) | Objective: maximize Precision (minimize wasted outreach)

1. Hyperparameter Search

RandomizedSearchCV (5-fold stratified CV), optimizing CV Precision. Search budgets: LR/HGB n_iter=50, RF n_iter=30 (total ~30 min). Results:

Model	Best CV Precision	Best Hyperparameters
Logistic Regression	0.7666	C=0.00108, penalty=l1, solver=liblinear, max_iter=1000
Random Forest	0.8018	max_depth=5, max_features=log2, min_samples_leaf=2, min_samples_split=10, n
HistGradientBoost (WINNER)	**0.8025**	**learning_rate=0.01432, max_iter=127, max_depth=10, l2_regularization=0.015**

HGB won by a narrow margin (0.8025 vs RF 0.8018) and is the most regularization-friendly; it was selected as the final model and calibrated in Task 4.

2. Diagnostics (Bias / Variance)

Learning curve on tuned HGB: train precision 0.8401 +/- 0.0038 vs validation 0.8012 +/- 0.0082 (gap 0.0389) - a GOOD FIT (small gap, low variance). Validation curves over learning_rate, max_iter, max_depth and l2_regularization confirmed plateau regions with no severe overfitting. Curves saved to day4-learning-curve.png and day4-validation-curves.png.

3. Calibration & Threshold Selection

Isotonic calibration (5-fold) improved reliability: Brier 0.0949 -> 0.0918 (improvement 0.0031; perfect = 0, random = 0.25). Threshold tuned for business precision.

Threshold	Precision	Recall	F1	Spec	TP	FP	FN
0.324 (best F1)	0.6467	0.7997	0.7151	0.8615	1254	685	314
0.500 (default)	0.7638	0.6250	0.6875	0.9387	980	303	588
0.538	0.8002	0.5797	0.6723	0.9541	909	227	659
0.832 (OPTIMAL)	**0.9695**	**0.3036**	**0.4624**	**0.9970**	**476**	**15**	**1092**

Optimal threshold for the business goal (max precision with recall >= 0.3) is 0.832, delivering 96.95% precision. Compared to default 0.5: +32.27% precision, 670 fewer false alarms, at the cost of 778 more missed positives.

4. Final Performance (Hold-out Test Set)

Precision	0.9695	**Optimal threshold**	0.832
Recall	0.3036	**Calibration**	Isotonic (5-fold)
F1 Score	0.4624	**Calibrated Brier**	0.0918
Specificity	0.9970	**ROC AUC**	0.9213

Confusion matrix at threshold 0.832: TN=4930, FP=15, FN=1092, TP=476. The saved artifact (day4-final-pipeline.joblib) captures preprocessing + feature engineering + selection + the HGB model, and is loaded at inference with the optimal threshold from day4-threshold-info.json. Expected production behavior: high-precision, low-recall outreach targeting only the most confident high earners.

Deliverables: day4-soln.ipynb | day4-final-pipeline.joblib | this report (day4-tuning-report.pdf)